

The empirical recurrence for OEIS A251804, recovered from its tail

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Abstract

OEIS A251804 records “Empirical recurrence of order 52” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 1024$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the whole sequence, so the recurrence holds from $n = 53$ — every index at which it can be written down. Its last coefficient is $c_{52} = -10319560704 \neq 0$, so no further tail satisfies anything shorter; any order-52 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A251804 is “Number of $(4+1)X(n+1)$ 0..3 arrays with no $2X2$ subblock having $x_{11}-x_{00}$ less than $x_{10}-x_{01}$.”. It has offset 1 and begins

216912, 18577404, 873945739, 26787273856, 594938582014, 10280556043804, 145329837643171, 174369221597120

The entry states:

Empirical recurrence of order 52 (see link above)

As of the “Last modified” line on the live entry (Jun 02 2025 10:32:38, revision 6) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 1024$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 1024$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Here the stated order is already the minimal order of the sequence itself, so no tail has to be taken.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 2048$ exact terms from $n = 1$ on, it returns order 52 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{52} c_i a(n-i),$$

c_1	90	c_{14}	−2589839087854120	c_{27}	289984594520728052024	c_{40}	−26418698816814
c_2	−3955	c_{15}	10695150716750224	c_{28}	−400021685064053046536	c_{41}	11365161484903
c_3	113106	c_{16}	−40147585537387510	c_{29}	509434841834571863792	c_{42}	−4359229104962
c_4	−2367232	c_{17}	137428864159554460	c_{30}	−598635461986089215576	c_{43}	14793471376646
c_5	38660016	c_{18}	−430148213321163554	c_{31}	648599315683307944688	c_{44}	−4399886290263
c_6	−512969848	c_{19}	1233881975749472428	c_{32}	−647285052592554161681	c_{45}	1133359848068
c_7	5685524976	c_{20}	−3249971374277096072	c_{33}	594250240246157011034	c_{46}	−249022847282
c_8	−53709422284	c_{21}	7872824863284915472	c_{34}	−501096462983116223363	c_{47}	457426249983
c_9	439095897688	c_{22}	−17562864981556608584	c_{35}	387377675856847698578	c_{48}	−68326771713
c_{10}	−3143916846148	c_{23}	36118300101741527824	c_{36}	−273926211846902375352	c_{49}	79712013385
c_{11}	19903008788152	c_{24}	−68528796290913454508	c_{37}	176709881648742782144	c_{50}	−6812200009
c_{12}	−112270053087544	c_{25}	120026972143339915864	c_{38}	−103668500101126228800	c_{51}	3792438558
c_{13}	567908765272720	c_{26}	−194130699375261936932	c_{39}	55102476520685444480	c_{52}	−103195607

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 53$, and it is the recurrence OEIS A251804 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 1$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{52} - c_1x^{52-1} - \dots - c_{52}$. Its constant term is $-c_{52} = 10319560704$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 52 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 52, so $\deg q = 52$, and it is monic. It holds from some index t on. From $\max(t, 1)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 52, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 13 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 52, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -10319560704 .

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-52 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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